

FreqST: Frequency-Domain Temporal Channel Stacking for Camera-Motion-Robust Action Recognition

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Goal. We propose FreqST, a preprocessing transform that lets a standard 2D-CNN image classifier implicitly capture short-term motion from a single frame-shaped input, with the specific aim of being more robust to camera motion than existing single-frame temporal-encoding methods.

Method. FreqST samples a window of W consecutive grayscale frames from a video clip and computes a 1D discrete cosine transform (DCT) along the time axis independently at every pixel, keeping only the DC coefficient and two low-frequency AC coefficients. This yields a standard 3-channel, image-shaped tensor per window: channel 0 encodes average appearance over the window, and channels 1–2 encode slow and fast motion energy respectively. Because the output has the same shape as a single RGB frame, it can be fed directly into any unmodified 2D-CNN image classifier, at the cost of one linear transform — no extra parameters and **no architecture change**.

We evaluate this preprocessing on the **CATER** benchmark (Girdhar & Ramanan, ICLR 2020), task 2 (compositional action recognition, 301 classes), on both its **static-camera** and **camera-motion** splits. We train three standard 2D CNN backbones (TSN, TSM, MVFNet) end-to-end with FreqST as the sole input preprocessing, using the original CATER training codebase and its standard metric (mean average precision, mAP), and compare directly against previously published baseline numbers for the same backbones and splits.

Prior work & delta. Kim et al. (BMVC 2022) [1] showed that two simple pixel-reordering strategies — Time-Color (TC) Reordering and GrayST — let an off-the-shelf 2D CNN implicitly capture short-term motion, by stacking a handful of raw grayscale/RGB frames from different time steps as if they were color channels, substantially beating single-RGB-frame baselines. However, on CATER’s camera-motion split their gains shrink sharply relative to the static split (e.g., TC Reordering: 73.7 $\rightarrow$ 56.6 mAP) — attributed to raw pixel-stacking being unable to distinguish “the object moved” from “the camera panned.”

Our delta: FreqST replaces raw-pixel stacking with an orthonormal per-pixel DCT over a full W -frame window (e.g. $W=8$), keeping 3 coefficients. In TC Reordering/GrayST, every channel is a raw snapshot frame — any appearance/motion split is incidental, not explicit. FreqST’s channels are a genuine decomposition instead: channel 0 is average appearance, channels 1–2 are motion energy, by construction. This also lets FreqST draw on 8 frames of context versus TC/GrayST’s 3, at the same channel budget — which we hypothesize helps separate a camera pan’s high-frequency background texture from the tracked object’s coherent motion, exactly what CATER’s camera-motion split tests.

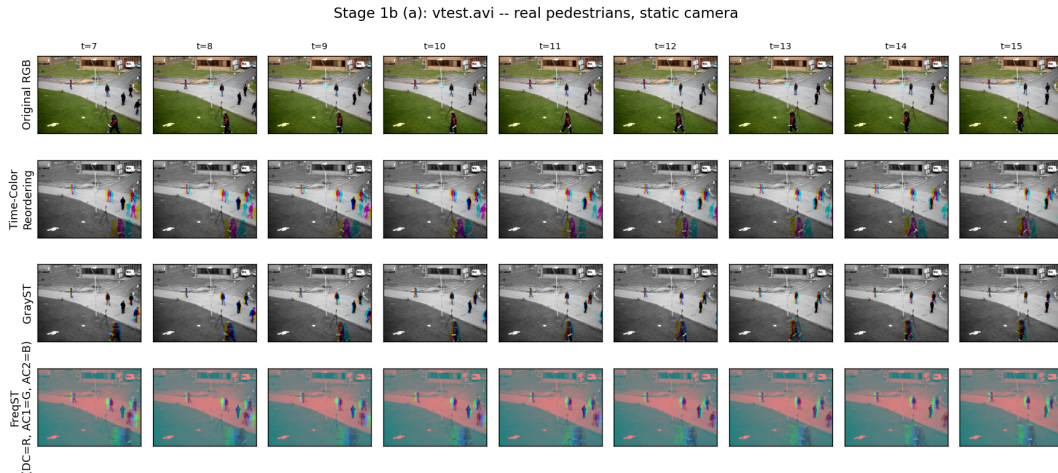


Fig. 1 — Same 8-frame real clip, static camera. Top to bottom: original frames; the two existing methods (Time-Color Reordering, GrayST — raw pixels/grayscale frames stacked as channels); FreqST (ours — DC/AC1/AC2 as R/G/B). In TC/GrayST colour appears only incidentally, wherever raw snapshots differ; FreqST’s channels are an explicit, dedicated appearance/motion split by construction.

Resources. This project needs **GPU compute** and about **35 GB of storage** for the CATER video and extracted-frame data.

Challenges & feasibility. The main open risk is generalization: CATER’s rendered scenes and scripted camera motion are a proxy for real footage, so the advantage may not transfer as cleanly with real backgrounds, occlusion, compression, and a pretrained backbone. A second risk is that any gain could be backbone- or dataset-specific — why we test three architectures (TSN, TSM, MVFNet) on both splits rather than one configuration. If FreqST underperforms, that is itself a useful, well-scoped negative result. On timeline: we start with a reduced-schedule pilot (fewer epochs, a data subset) on both splits, fast to run and telling us whether the full-schedule study across all three backbones is warranted — keeping the most expensive compute conditional on an early positive signal.

[1] Kim, K., Gowda, S. N., Mac Aodha, O., & Sevilla-Lara, L. (2022). *Capturing Temporal Information in a Single Frame: Channel Sampling Strategies for Action Recognition*. BMVC 2022.